Aspect	*Amazon Alexa / Alexa Skills Kit*	*Google Assistant / Google Cloud services*	*Apple Siri / App Intents*	*Microsoft Azure Speech + Bot Framework*	*SAM Voice Service*
Primary orientation	Consumer and smart-home assistant ecosystem with developer-created skills	Consumer assistant ecosystem plus commercial cloud speech/NLP services	Device-integrated assistant focused on Apple ecosystem and app-level actions	Commercial cloud platform for speech, bots and conversational services	Assistive voice-service architecture for older adults and care-service contexts
Openness and modularity	Skills can define custom interaction models, but the core assistant platform remains closed	Cloud APIs are modular, but Google Assistant custom conversational actions have been sunset	Developers expose app capabilities through SiriKit/App Intents, but assistant behaviour remains platform-controlled	Modular SDKs and cloud services for speech and bots	Fully architecture-defined modular service stack: ASR, NLU, dialogue management, NLG/TTS, gateway, broker and persistence services
Deployment control	Mainly dependent on Amazon-managed Alexa infrastructure and certification process	Cloud-managed services; Assistant-based custom conversational deployment is limited after the Actions sunset	Bound to Apple devices and system-level Siri/App Intents integration	Azure cloud deployment, with some container/on-premises options for speech services	Public or private cloud deployment under control of the service provider
Scalability model	Platform-level scalability handled by Amazon	Platform/cloud scalability handled by Google services	Device/ecosystem-level scalability handled by Apple	Cloud scalability through Azure services and containers	Service-level scaling through containerised microservices, Kubernetes and load balancing

Privacy and data locality	Privacy governed by Amazon platform policies; skill developers control only their own backend data	Privacy and data processing governed by Google service configuration and policies	Strong device/ecosystem privacy model, but limited provider control over Siri internals	Enterprise-grade cloud controls; speech containers can support more controlled deployments	Data flow and retention policies can be governed by the deploying care-service organisation
Personalisation and adaptation	Skill-level personalisation possible; limited control over core ASR/TTS and assistant behaviour	Speech adaptation and cloud customisation are available, but not a dedicated older-adult assistant architecture	App-level intents and shortcuts; limited customisation of core Siri voice/dialogue models	Custom speech and bot configuration available; not specifically oriented to older adults by default	Speech and dialogue models can be adapted to older adults, accents, care routines and local-service needs
Technology transparency	Core technologies are proprietary	Core cloud models and Assistant behaviour are proprietary	Core Siri technologies are proprietary	Commercial APIs and SDKs; some components configurable but platform-dependent	Open architecture based on replaceable components and reproducible deployment
Suitability for older-adult care services	Mature ecosystem, but limited control over assistant behaviour, wake interaction and domain-specific model adaptation	Powerful cloud AI services, but Assistant extensibility has changed and may constrain long-term care-service design	Strong device integration, but less suitable for provider-controlled standalone assistive devices	Strong enterprise option, but may increase vendor dependence and cost	Designed specifically for assistive services, provider-controlled operation and care-context adaptation